NuMI proton-beam position on target versus 2x2 / Mx2 rates: Run 1 proof of concept and target-scan feasibility

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Builds on the `beam_quality` package of G. Ingratta and B. Howard (York U.)

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code: `joshuabarrow221/arcube_nearline`, branch `claude/project-thread-e9bc00`, dir `beam_position_scan/`

Confidence tags: [CERTAIN] = measured here from data or exact arithmetic; [LIKELY] = strong inference from data plus standard physics; [GUESSING] = filling a gap, must be checked.

1 Executive summary

1. **Run 1 cannot pinpoint an "optimal" beam position, and the batch-split test is expected to come out null.** Batch 1 is displaced by -0.314 mm in x (RMS 0.029 mm spill to spill) and -0.055 mm in y relative to batches 2–6 [CERTAIN]. Against a 7.4 mm-wide target fin and a $\sigma_x = 1.48$ mm beam this changes the fraction of protons hitting the fin by only +0.24% if the beam sits at the BPM-coordinate origin, and by +1.74% / -0.89% if the fin centre is 1 mm to either side [LIKELY]. Run 1 should yield $\sim 6.5 \times 10^5$ selected in-spill light flashes (projected from the 19-file rate, [LIKELY]), enough to see a 1.0% batch-1/batches-2–6 difference at 3σ. The batch split can therefore only flag a beam that was $\gtrsim 1$ mm off the fin centre in x. It cannot locate an optimum. Measured on 7,868 spills from 19 files: $R_1/R_{2\text{–}6} = 0.992 \pm 0.026$ for all flashes and 1.000 ± 0.030 for flashes lighting $\geq$2 TPCs. Both are null, as predicted [CERTAIN].
2. **Halving or quartering the batch "ellipses" adds nothing in x.** The beam is extremely steady in x (spill RMS 0.025 mm), so splitting each batch population at its median gives a lever arm of only -0.055 mm (batch 1) and -0.092 mm (batches 2–6) [CERTAIN]. In y the halves differ by -0.293 mm, but the fin is 63 mm tall, so interception has no y dependence there. Any y effect would come from horn focusing alone, which is much weaker [LIKELY].
3. **The batch comparison is confounded by time order, and the bias is bigger than the signal.** Batch 1 is always first in the spill. Delayed "follower" flashes from earlier interactions ($\tau \approx 3.6\,\mu$s, consistent with stopped-muon decays) pile into the later batches and never into batch 1. Measured from the post-spill tail, this biases $R_1/R_{2\text{–}6}$ by -2.1% for all flashes and -0.3% for flashes lighting $\geq$2 TPCs [LIKELY]. Both are larger than the $\sim$0.2% interception signal expected for a centred beam. No pile-up *loss* is seen: later batches have, if anything, more flashes after an earlier one (batch 6: 1.10 ± 0.06) (§8).
4. **A dedicated horizontal scan that reaches the fin edges is feasible with the 2x2 light system at** 100 kW**.** The measured in-spill flash rate is 0.295 flashes per 10^{13} POT ($\approx$1.50 per spill at 816 kW). That corresponds to $\approx$504 flashes per hour at 100 kW, about ten times the $\sim$50 neutrinos/hour assumed in the email thread. With 59 counts per point, a 7-point scan over $\pm$4.5 mm locates the fin centre to $\pm$0.25 mm (statistical), and about 400 counts per point gives $\pm$0.1 mm (toy MC). That is roughly 7 min and 48 min per point, respectively, using all flashes. A

plateau scan (±1 mm about nominal) does not converge even with 2000 counts per point (25% of toys find no maximum) [LIKELY].

5. **The detector can confirm the target centre. It is not competitive with the accelerator's own instruments for optimisation.** NuMI's beam-based alignment reaches ∼0.03 mm with profile monitors and ∼0.26 mm with BPMs, and uses hadron and muon monitors with $10^{6–8}$ particles per spill [1]. The useful product of a 2x2 scan is an independent, neutrino-side check that the rate-weighted target centre agrees with the BPM centre.
6. **Run 1 was taken in reverse horn current (antineutrino mode)** [4], yet the `beam_quality` horn cut ($-202 < I < -196.4$ kA, documented as FHC) passes every Run 1 spill with $I \approx -200.5$ kA. The stripline sum evidently does not encode polarity. The package README's statement that the 2024 data are "nominal FHC running" is therefore probably wrong [LIKELY]. This does not affect the position analysis, but it should be fixed before the horn cut is used to classify polarity.

2 Question, hypotheses, and what was tested

The request: determine how the proton-beam position on the NuMI target affects the rates seen in 2x2 and Mx2, whether "optimal" positions can be pinpointed, and whether past data can confirm the approach. The proposed proof of concept uses batch 1 ("not as kicked") and batches 2–6 as two simultaneous beam positions, subdivides them if statistics allow, and extrapolates to ∼100 kW with ∼1 h per position.

Where the premise holds and where it does not.

- Using batches as simultaneous positions is a good idea. Every detector, live-time, purity and beam-intensity condition is shared, so only the position (and the time order, §8) differ. The data support this: the offset is real, stable and measured spill by spill (§4).
- I disagree that the neutrino-optimal position is where "true vertices are directly on axis." A sub-mm parallel displacement at the target moves the neutrino-beam centre at 1 km by the same sub-mm amount. That is invisible against a beam several metres wide. What changes the rate is (a) the fraction of protons intercepted by the fin and (b) the horn-focusing acceptance for secondaries made off axis. Here's what I'd do instead: define the optimum operationally as the centre of the rate-versus-position response, found from its edges. The risk in your approach is chasing a maximum on a plateau that is flat to <1% over ±1 mm (§5).
- I disagree that rock-muon rates are an uncontroversial proxy for the neutrino rate in this context. Rock muons are weighted toward higher neutrino energies, whose parents are less affected by horn focusing than the flux peak. Interception changes affect both equally, so an edge scan works with either. For focusing-driven changes near the optimum, the two proxies can disagree [LIKELY]. Here's what I'd do instead: report the fully contained, rock-muon-like (≥ 4 TPCs lit), and all-flash rates separately and require them to agree. The risk in your approach is that a combined "neutrino-equivalent rate" hides a proxy-dependent optimum.
- "∼50 neutrinos/hour/position" undersells the detector for this purpose. The quantity that matters is any in-spill activity whose rate is proportional to the flux: light flashes from rock muons and neutrino interactions in and around the argon. Measured: ≈504/h at 100 kW [LIKELY] (one-file rate, §6).

3 Data and access

- **Beam:** IFBeam per-spill records on TCLK $A9 for the BPM arrays `E:HP121[]`, `E:VP121[]`, `E:HPTGT[]`, `E:VPTGT[]` and the intensities `E:HITGT[]`, `E:VITGT[]`. Element 0 is the auto-tune average and elements 1–6 are batches. The two existing `get_data.cpp` exports were used: 2024-07-09 00:00–02:05 and 2024-07-12 00:01–01:01 CDT, with 8,443 spills, 8,406 of them goodbeam, and 4.28×10^{17} POT. The scalar spellings (e.g. `E:HPTGT`) return HTTP 200 with *zero rows*. Code that does not check for empty rows silently gets nothing [CERTAIN].
- **2x2:** Run 1 merged charge+light FLOW, reflow v11 (flow v1.5.0), streamed by HTTP byte ranges from https://portal.nersc.gov/project/dune/data/2x2/reflows/v11/flow/beam/. Only `light/events`, `light/t_ns`, `light/sum_hits` and the event–hit references are read; the 8 GB of waveforms per file are not. The Run 1 RunsDB (`mx2x2runs_v0.2.sqlite`) gives 2.2×10^{19} POT with CRS and LRS both recorded (3.0×10^{19} LRS only).
- **Mx2:** MINERvA DSTs are on the portal (`data/2x2/Mx2/dst/`: 502 files, ∼374 MB each, runs 50000–50018). Each file has one tree, `minerva`, with one entry per gate (gates 1.1997 s apart) and per-hit `hit_time`, `hit_module`, `hit_plane`, `hit_pe`. That is enough to assign front-plane hits to batches. File-name times are UTC [CERTAIN]. They were not analysed here; see §9.
- **Access constraint:** neither portal.nersc.gov nor the IFBeam API is reachable from the cloud machine that wrote this report. All fetches ran on the requester's laptop. The full-Run 1 extraction should run at NERSC against the files directly.

4 Beam side: is batch 1 really somewhere else?

Per-batch positions at the target z were obtained by the same linear extrapolation from the 121 and TGT BPM stations used by the NOvA `BpmProjection` port in `get_data.cpp`. Fig. 1 and Table 1 summarise them.

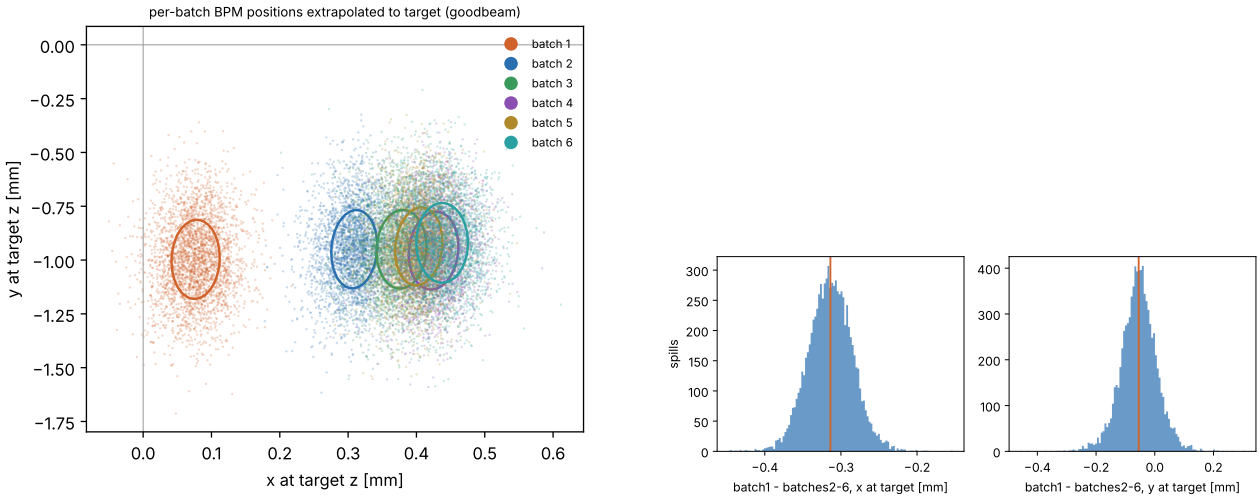


Figure 1: Left: per-batch BPM positions extrapolated to the target, goodbeam spills, both windows (ellipses: 1σ). Right: per-spill offset of batch 1 from the intensity-weighted mean of batches 2–6.

Findings [CERTAIN]:

- Batch 1 minus batches 2–6: $\Delta x = -0.314 \pm 0.0003$ mm (RMS 0.029), $\Delta y = -0.055$ mm (RMS 0.062). The offset is constant to ± 0.01 mm across 10-minute blocks spanning both days.

batch	1	2	3	4	5	6
$\langle x \rangle$ at target [mm]	+0.077	+0.309	+0.378	+0.425	+0.404	+0.437
intensity share [%]	16.29	16.60	16.62	17.01	16.93	16.55

Table 1: Per-batch means. Statistical errors on the means are <0.001 mm.

- Batches 2–6 are not one position either. They drift monotonically by ∼0.13 mm in x from batch 2 to batch 6, consistent with a sagging kicker flat-top.
- Batch 1 carries less beam (16.29% of the spill against ∼16.6–17.0%), so rates must be per POT *per batch*.
- Spill-to-spill jitter is tiny in x (RMS 0.025 mm) and larger in y (0.18 mm). Beam width from the multiwire: $\sigma_x = 1.48$, $\sigma_y = 1.48$ mm.

The batch offset is *larger* than the jitter, so it is the only usable x lever arm in normal running. It is a fifth of the beam width and a twentieth of the fin width.

5 Rate model

The only position dependence computable without a beamline simulation is interception. For a Gaussian beam of RMS σ centred at x_0 from the centre of a fin of full width w,

$$f(x_0) = \Phi\left(\tfrac{w/2-x_0}{\sigma}\right) - \Phi\left(\tfrac{-w/2-x_0}{\sigma}\right), \qquad \frac{f(x_0)}{f(0)} \simeq 1 - k\,x_0^2, \quad k = \frac{w/\sigma^3\,\phi(w/2\sigma)}{2f(0)} = 0.0203\ \text{mm}^{-2}.$$

Protons that miss the fin are given a relative neutrino yield $\epsilon = 0.2$ ([GUESSING]; results below are insensitive to it). With $w = 7.4$ mm (the NOvA-era ME fin [3]; the 1 MW-era fin width was not confirmed, [GUESSING]) and $\sigma = 1.48$ mm the relative rate is 0.982 at 1 mm, 0.909 at 2 mm, 0.753 at 3 mm and 0.440 at 4.5 mm (Fig. 2). For scale, the 2005 NuMI beam-based alignment quotes ∼2% spectral distortions for ∼1 mm target/baffle offsets in the low-energy configuration [1]. Focusing effects of that size are not in this model. They need g4numi derivatives for the present configuration, which would replace the free quadratic term in `target_model.relative_rate`.

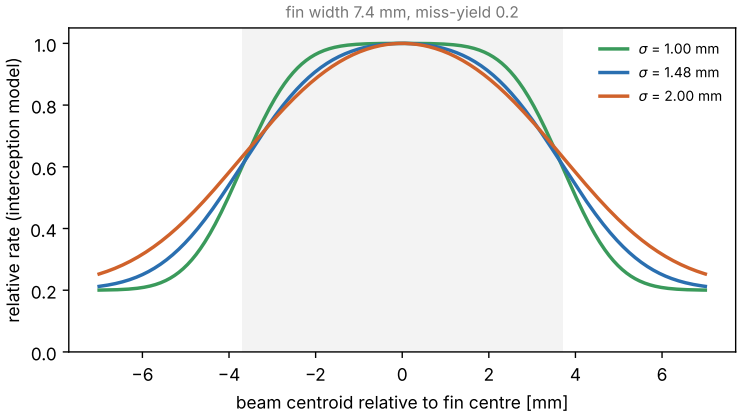


Figure 2: Relative rate versus beam offset from the fin centre (interception model). The shaded band is the fin.

6 Detector side: per-batch light-flash rates

Why light, not charge. CRS timestamps are smeared by drift times of up to ∼200 µs, so a charge cluster cannot be assigned to a 1.58 µs batch without an external t_0. The LRS digitises a 16 µs window opened by the beam trigger at 16 ns per sample, so each pulse's time relative to the trigger is known to ∼10–20 ns [CERTAIN]. The spill appears as a ≈9.5 µs plateau starting ≈1317 ns after the trigger (Fig. 3).

Selection.

1. Beam triggers are `light/events` with `trig_type`=1. In the file checked, 454 such triggers matched 455 IFBeam spills; this mapping is not documented ([LIKELY]).
2. Spill matching uses the light DAQ unix time: trigger − \$A9 = 13.0 ms, RMS 2.9 ms. Spills with |residual| > 10 ms are dropped as timing drifts, per the requirement to remove rather than correct such events.
3. Summed-channel hits (`sum_hits`) are timed with `busy_ns`+`ns_spline`. The `ns` field is the *event* time stored as float64 (∼256 ns precision at 1.7×10^{18}), not the hit time [CERTAIN]. Hits within 80 ns are merged across all TPCs into flashes, so a through-going muon counts once. Flashes need ≥ 3 hits.
4. The batch-1 start is fitted from the data; six nominal batches of 84 RF buckets (1581.8 ns) follow. Guard bands of 50 ns (100 ns around batch 1) are removed and the POT is scaled by the surviving fraction of each batch.

Result (19 files: 2024-07-09 23:54–02:05 and 2024-07-12 00:00–01:00 CDT, runs 50017/50018). 9,320 beam triggers; 282 unmatched, 14 dropped for timing, 1,143 duplicate matches and 18 not good-beam, leaving 7,868 spills and 11,937 selected flashes. Batch 1 / batches 2–6 per-POT rate ratio:

$$R_1/R_{2\text{–}6} = 0.992 \pm 0.026 \quad (p = 0.75),$$

with 1,732 and 9,559 flashes. The 3σ reach of this subsample is 8%, an order of magnitude above the ∼0.2% expected for a centred beam. In the interception model ($\sigma = 1.48$ mm, $w = 7.4$ mm) a ratio this far from unity needs the fin centre $\gtrsim$3 mm from the beam, so this sample excludes only a gross miss [LIKELY]. The full Run 1 sample (reach ≈1%) would reach ≈1 mm. Requiring ≥2 lit TPCs (which suppresses single-TPC followers) gives 1.000 ± 0.030 from 8,613 flashes. Correcting each for the follower bias gives 1.013 and 1.003, respectively.

The batch edges are visible in the data. The flash-time distribution (Fig. 3) has narrow dips at the fitted batch boundaries: the empty RF buckets between Booster batches. So the batch assignment is checked by the data themselves, not only by the 84-bucket arithmetic [CERTAIN].

Data defect: duplicated light events. Two of the 19 v11 files (`packet-0050017-2024_07_09_01_14_40` and `packet-0050018-2024_07_12_00_00_19`) contain repeated light triggers: 687 and 456 beam triggers, respectively, share an IFBeam spill with another trigger. All 1,143 such triggers were dropped, together with the spills they matched, rather than guessing which copy is real. This should go to the reflow maintainers [CERTAIN].

7 Sensitivity and extrapolation

Batch split, Run 1. With a fraction $\varphi = 1/6$ of the POT in batch 1, a fractional difference δ is seen at $Z\sigma$ when $N_{\text{tot}} \geq Z^2\delta^{-2}(\varphi^{-1} + (1-\varphi)^{-1}) = 7.2Z^2/\delta^2$ (verified by toys in `tests/`). The measured

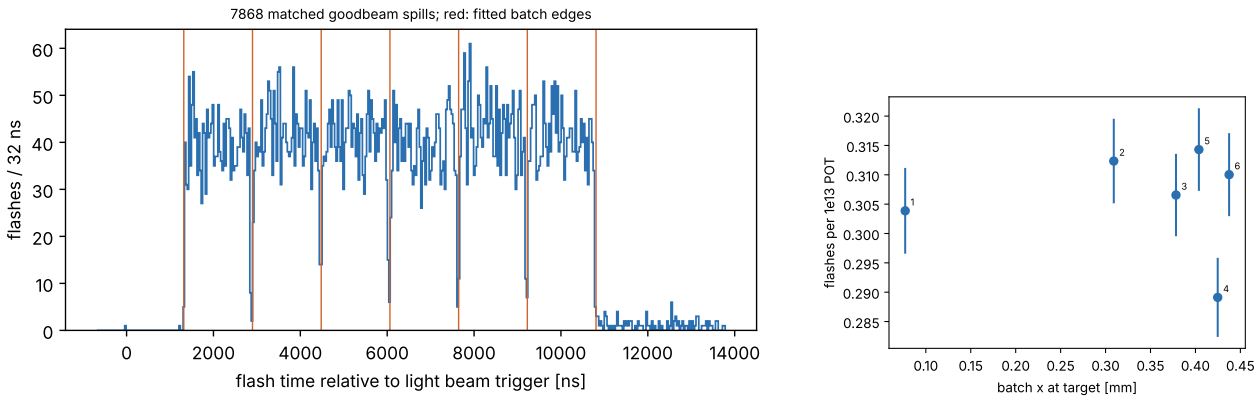


Figure 3: Left: flash times relative to the light beam trigger, with the fitted batch edges. Right: per-POT flash rate per batch against that batch's mean x at the target.

flash rate projects to $\approx 6.5 \times 10^5$ selected flashes for the 2.2×10^{19} POT with CRS+LRS live, giving a 1.0% 3σ reach (Fig. 4). The model predicts $|\delta| \gtrsim 1\%$ only if the fin centre is $\gtrsim$1 mm from the beam in x. A further 0.3% per-batch POT-share uncertainty (BPM intensities used as weights) is included in the dashed curve.

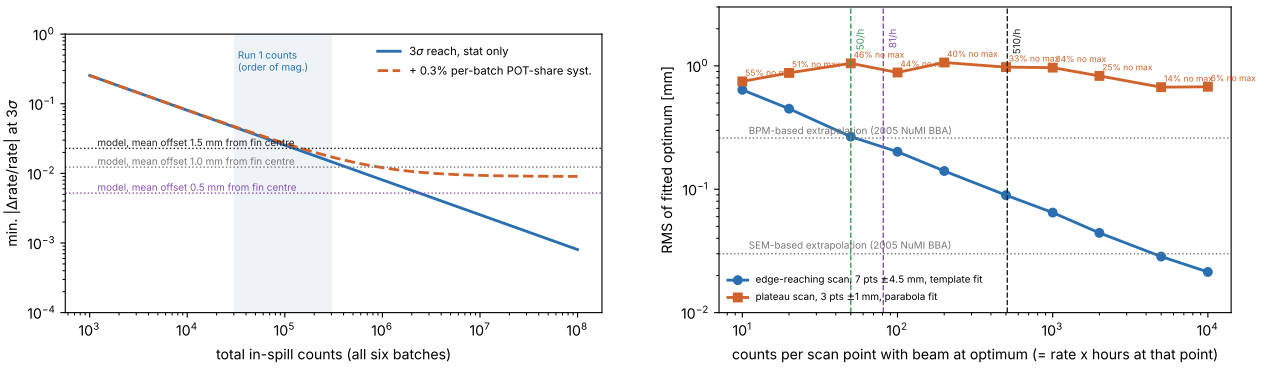


Figure 4: Left: 3σ reach of the batch-1 versus batches-2–6 comparison against total counts, with model predictions for different fin-centre offsets. Right: toy-MC precision on the fitted optimum against counts per scan point. The dashed verticals mark assumed hourly rates at 100 kW; the measured rate is ≈504/h.

Dedicated scan at 100 kW. Detector rates scale with POT. At 100 kW with a 1.2 s cycle and 46 spills per 60.46 s supercycle, the flash yield is ≈504/h for all selected flashes and ≈79/h for flashes lighting ≥4 TPCs (rock-muon-like). Toy-MC results for a 7-point scan over ±4.5 mm with a template fit of the known shape:

target precision on fin centre	0.5 mm	0.25 mm	0.1 mm
counts per point (at optimum)	16	59	400
time per point, all flashes (≈504/h)	2 min	7 min	48 min
time per point, ≥4 TPCs (≈79/h)	12 min	45 min	5.1 h
time per point, contained ν (∼50/h, email estimate)	19 min	70 min	8.0 h

Fitting with a deliberately wrong template (σ 30% too large and $\epsilon = 0$) shifts the centre by only -0.06 mm, because a symmetric shape error does not move the centre [LIKELY]. A ±1 mm plateau

scan fails even at 2000 counts per point: 25% of toys have no maximum, and the rest scatter by 0.83 mm. **One hour per position at 100 kW is enough if, and only if, the scan reaches the fin edges.** Whether running the beam onto the fin edge at 100 kW is acceptable (baffle and fin-edge heating) is the accelerator's call.

8 Systematics and caveats

- **Time-order bias in the batch split.** Two opposite effects are possible: masking (an earlier flash hides a later one) and followers (an earlier interaction adds a delayed flash). The data show followers dominating. The post-spill tail decays with $\tau \approx 3.6\,\mu$s, and the steady-state follower fraction is 3.9% of in-spill flashes (1.0% with $\geq$2 TPCs). The expected follower share rises from batch 1 to batch 6, biasing $R_1/R_{2\text{–}6}$ by -2.1% (-0.3%). The tail fit uses only ∼100 flashes, so these corrections are uncertain at the ∼30–50% level [LIKELY]. `detector_rates.follower_bias` implements the estimate. The scan is not affected, because position does not correlate with time order there.
- **Per-batch POT.** The batch shares come from BPM intensities, not toroids. A 0.3% share error is already comparable to the expected signal.
- **Trigger identity.** `trig_type`=1 was matched to IFBeam counts in one file only. Confirm it with the LRS configuration.
- **Horn polarity labelling** (summary item 6).
- **Vertical plane.** The fin is tall, so a vertical scan has no edges within a safe range. Any vertical optimisation relies on focusing effects of order 1%/mm.
- **Fin width at 1 MW.** The 7.4 mm value is for the NOvA-era ME target. The centre found by an edge scan does not depend on it; the predicted batch-split size does.

9 Recommendations and next steps

1. Tell the accelerator team: the 2x2 can provide an independent, neutrino-side measurement of the horizontal target centre to ∼0.1–0.25 mm in a 7-point, ±4.5 mm scan at 100 kW with 7 min–48 min per point (all flashes) or 45 min–5.1 h (rock-muon-like only). That holds if they can sit at those offsets, and the scan time is driven by their safety limits rather than our statistics. Do not offer to find a rate maximum on the plateau.
2. Ask for the scan's per-setpoint BPM targets and timestamps in IFBeam (the existing `[]` devices suffice). No new ACNET parameters are needed for our side.
3. Run `extract_lrs_flashes.py` on all Run 1 v11 files *at NERSC* (local filesystem, no streaming), fetch per-batch IFBeam data for the same span, and run `batch_rates.py`. Include the pile-up study before quoting a ratio.
4. Mx2: write a DST reader that emits (spill time, hit time relative to gate) for the front planes, and feed it to `detector_rates`. MINERvA timing (∼ns) resolves batches as well. It adds an independent rock-muon sample with different acceptance.
5. Obtain g4numi derivatives of the 2x2 flash and contained-ν rates with respect to beam x, y and angle for the current target and horns. They replace the interception-only model before any scan decision.

10 Reproduction

See `beam_position_scan/README.md`. In short:

```
python3 scripts/analyze_ifbeam_batches.py STEM1 STEM2 --out results/ifbeam
python3 scripts/sensitivity_study.py --sigma 1.48 --width 7.4
python3 scripts/extract_lrs_flashes.py --out flashes/ <FLOW files or URLs>
python3 scripts/batch_rates.py --flashes flashes/*.npz --ifbeam STEM1 STEM2 \
        --out results/batch_rates
python3 scripts/make_report_numbers.py && cd report && pdflatex beam_position_scan.tex
```

References

[1] R. Zwaska *et al.*, "Beam-based alignment of the NuMI target station components at FNAL," NIM A 568 (2006) 548, arXiv:physics/0609106.

[2] P. Adamson *et al.*, "The NuMI neutrino beam," NIM A 806 (2016) 279, arXiv:1507.06690.

[3] K. Ammigan *et al.*, "Study of a new target design with an additional horn for NuMI beam," arXiv:1710.03699 (NOvA ME fins: 7.4 mm wide, 63 mm tall, 24 mm long; 1.3 mm RMS spot).

[4] DUNE 2x2 physics talk, NuFact 2024, indico.fnal.gov/event/63406 ($\sim 10^{19}$ POT in antineutrino mode, ∼10k interactions/day).